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Studierichting: Informatica
Oefeningenreeks 3G nr 7.5

$f(x) = (x-1)(x-2)(x-3) = x^3 - 6x^2 + 11x - 6$

$\operatorname{dom} f : x \in \mathbb{R}$

$\Rightarrow$nulpunten: $x = 1, x = 2, x = 3$

Assymptoten:

Geen VA want $\operatorname{dom} f(x) = \mathbb{R}$

HA: $\lim\limits_{x\to\infty} f(x) = \infty \Rightarrow$geen HA

SA: $\lim\limits_{x\to\infty} \frac{f(x)}{x} = \infty \Rightarrow$geen SA

$f'(x) = 3x^2 - 12x + 11$

$\Rightarrow$nulpunten: $3x^2 - 12x + 11 = 0 \Rightarrow D = 12$

$x_1 = 2 + \frac{\sqrt{3}}{3}$

$x_2 = 2 - \frac{\sqrt{3}}{3}$

$f''(x) = 6x - 12$

$\Rightarrow$nulpunten: $6x - 12 = 0 \Rightarrow x = 2$

x		1		$2-\frac{\sqrt{3}}{3}$		2		$2+\frac{\sqrt{3}}{3}$		3	
$f(x)$	$-$	0	$+$	$+$	$+$	0	$-$	$-$	$-$	0	$+$
$f'(x)$	$+$	$+$	$+$	0	$-$	$-$	$-$	0	$+$	$+$	$+$
$f''(x)$	$-$	$-$	$-$	$-$	$-$	0	$+$	$+$	$+$	$+$	$+$
$f(x)$	$\nearrow$	$\nearrow$	$\nearrow$	$M\left(\frac{2\sqrt{3}}{9}\right)$	$\searrow$	$\searrow$	$\searrow$	$m\left(\frac{-2\sqrt{3}}{9}\right)$	$\nearrow$	$\nearrow$	$\nearrow$
	$\frown$	$\frown$	$\frown$	$\frown$	$\frown$	0	$\smile$	$\smile$	$\smile$	$\smile$	$\smile$

Tekening:

References

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